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IF351 DATABASE SYSTEMS

05 DATA DEFINITION LANGUAGE (DDL) AND DATA MANIPULATION LANGUAGE (DML) STATEMENTS



REVIEW:

THE RELATIONAL DATA MODEL, RELATIONAL DATABASE CONSTRAINTS, AND RELATIONAL DATABASE DESIGN

- Relational Data Model
- Relational Database Constraints
- ER-to-Relational Mapping Algorithm

COURSE SUB LEARNING OUTCOMES (SUB-CLO)

- Sub-CLO 5
 - Students are able to use D D L statements to create a database (C3)
- Sub-CLO 6
 - Students are able to use DML statements to create a database (C3)

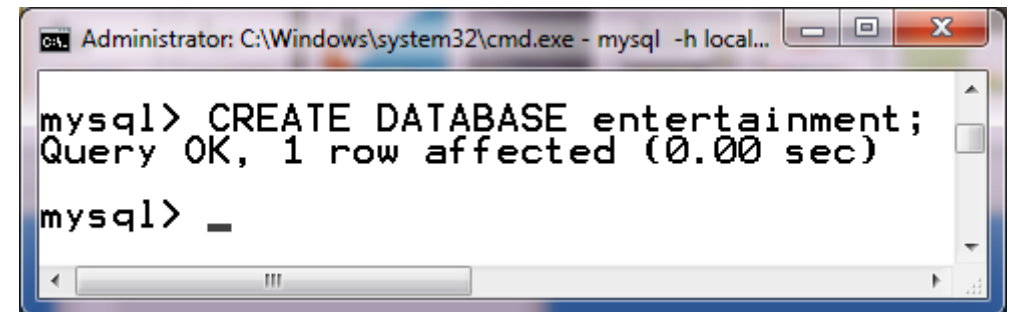
OUTLINE

- Creating, Showing, & Deleting Databases
- Selecting the Current Database
- Data Types of Columns
- Creating, Showing, & Deleting Tables
- Populating Tables with Data
- Querying Tables
- Updating & Deleting Rows
- Copying Tables
- Populating a Table with Rows from Another Table

CREATING DATABASES

- A database acts as a container for a set of tables
- For MySQL, each table must be created within an existing database
- Therefore, when you want to build a table, you first need to create a database

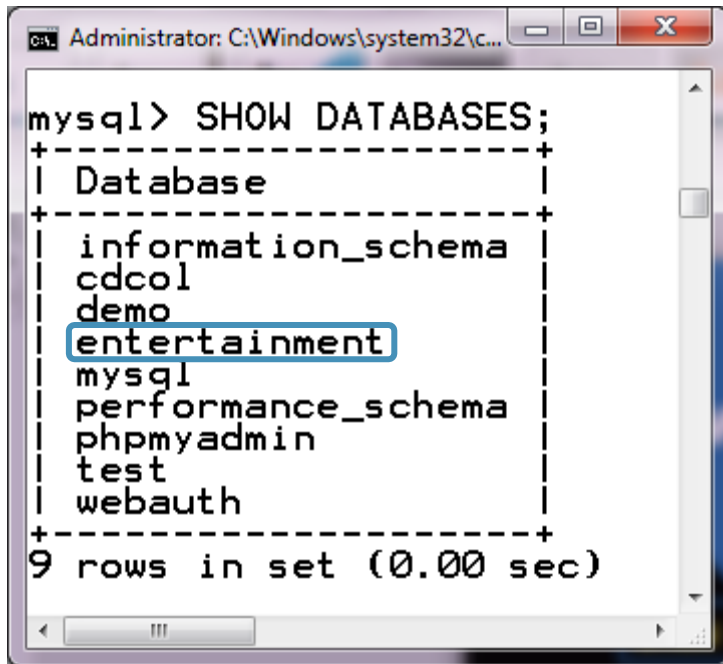
```
CREATE DATABASE database_name;
```



```
Administrator: C:\Windows\system32\cmd.exe - mysql -h local...  
mysql> CREATE DATABASE entertainment;  
Query OK, 1 row affected (0.00 sec)  
mysql> _
```

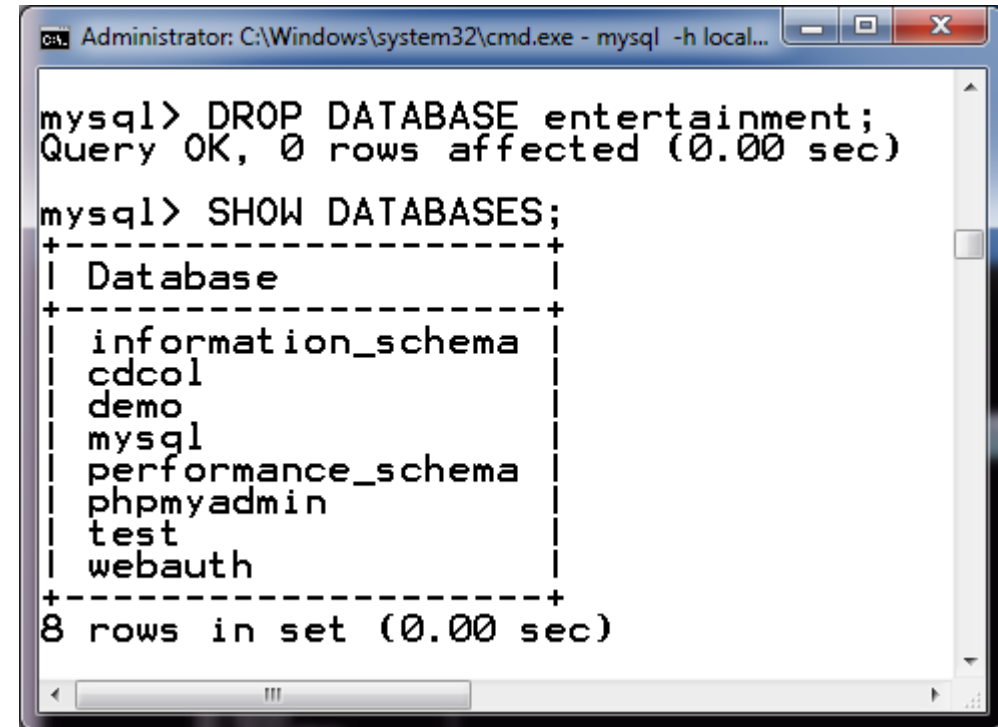
SHOWING & DELETING DATABASES

SHOW DATABASES;



```
Administrator: C:\Windows\system32\cmd.exe
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| cdcol          |
| demo           |
| entertainment    |
| mysql          |
| performance_schema |
| phpmyadmin      |
| test           |
| webauth        |
+-----+
9 rows in set (0.00 sec)
```

DROP DATABASE database_name;



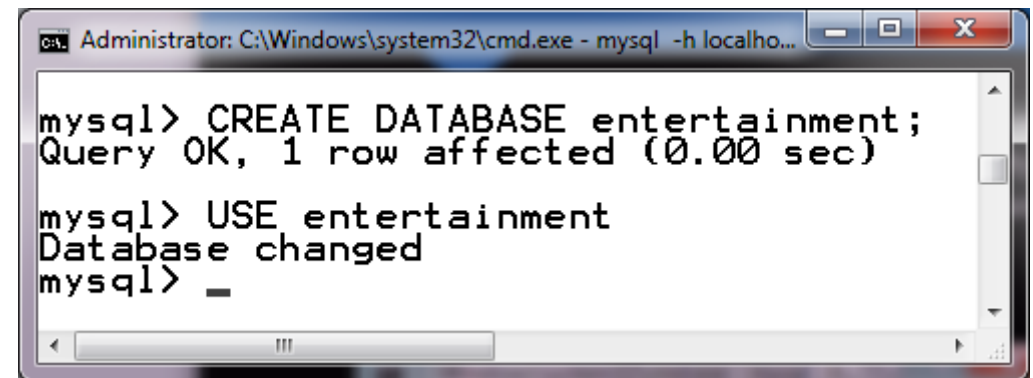
```
Administrator: C:\Windows\system32\cmd.exe - mysql -h local...
mysql> DROP DATABASE entertainment;
Query OK, 0 rows affected (0.00 sec)

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| cdcol          |
| demo           |
| mysql          |
| performance_schema |
| phpmyadmin      |
| test           |
| webauth        |
+-----+
8 rows in set (0.00 sec)
```

SELECTING THE CURRENT DATABASE

USE database_name

- When a user has opened a connection with MySQL and wants, for example, to create new tables or query existing tables, the user must specify the database he wants to work with



```
Administrator: C:\Windows\system32\cmd.exe - mysql -h localho...  
mysql> CREATE DATABASE entertainment;  
Query OK, 1 row affected (0.00 sec)  
  
mysql> USE entertainment  
Database changed  
mysql> _
```

DATA TYPES OF COLUMNS: INTEGER DATA TYPES

Integer Literal	Range
TINYINT	-2^7 up to and including $+2^7-1$
SMALLINT	-2^{15} up to and including $+2^{15}-1$
MEDIUMINT	-2^{23} up to and including $+2^{23}-1$
INTEGER	-2^{31} up to and including $+2^{31}-1$
BIGINT	-2^{63} up to and including $+2^{63}-1$

Original Data Type	Synonyms
TINYINT	INT1
SMALLINT	INT2
MEDIUMINT	INT3, MIDDLEINT
INTEGER	INT, INT4
BIGINT	INT8

DATA TYPES OF COLUMNS

- Decimal Data Types

- DECIMAL
- DEC
- NUMERIC
- FIXED

DECIMAL(8,2)

- Float Data Types

- FLOAT
- FLOAT4
- REAL
- DOUBLE

DATA TYPES OF COLUMNS: ALPHANUMERIC DATA TYPES

Alphanumeric Data Type	Maximum Length
CHAR	(2 ⁸ -1) characters
VARCHAR	(2 ⁸ -1) characters (2 ¹⁶ -1) characters from version 5.03
LONG VARCHAR	(2 ²⁴ -1) characters
LONGTEXT	(2 ³² -1) characters

Alphanumeric Data Type	Synonyms
CHAR	CHARACTER NCHAR NATIONAL CHAR NATIONAL CHARACTER
VARCHAR	CHARVARYING CHARACTER VARYING NATIONAL VARCHAR NATIONAL CHAR VARYING NATIONAL CHARACTERVARYING TEXT TINYTEXT (maximum length = 255)
LONG VARCHAR	MEDIUMTEXT
LONGTEXT	

DATA TYPES OF COLUMNS

- Temporal Data Types
 - DATE
 - TIME
 - DATETIME
 - TIMESTAMP
 - YEAR
- Bit Data Type
- Blob Data Types
- Geometric Data Types

CREATING TABLES

- Databases in MySQL are made up of database objects
- The best-known and most important database object is probably the table

```
CREATE TABLE table_name(  
    column_name    column_data_type    null_specification    column_integrity_constraint ,  
    column_name    column_data_type    null_specification    column_integrity_constraint ,  
    ... ,  
    table_integrity_constraint ,  
    table_integrity_constraint ,  
    ...  
);
```

```
CREATE TABLE actors(  
    id            CHAR(5),  
    first_name    VARCHAR(15),  
    last_name     VARCHAR(20),  
    gender        CHAR(1),  
    birth_date    DATE  
);
```

CREATING TABLES: NULL SPECIFICATION

- **NOT NULL** indicates which columns cannot contain NULL values
- **NULL:NULL** values are allowed
- Not including a NULL specification is equal to specifying NULL

```
CREATE TABLE actors(  
    id          CHAR(5) NOT NULL,  
    first_name  VARCHAR(15) NOT NULL,  
    last_name   VARCHAR(20),  
    gender      CHAR(1) NOT NULL,  
    birth_date  DATE  
);
```

CREATING TABLES: SPECIFYING INTEGRITY CONSTRAINTS

PRIMARY KEYS

- As column integrity constraints

```
CREATE TABLE actors(  
    id          CHAR(5) PRIMARY KEY,  
    first_name  VARCHAR(15) NOT NULL,  
    last_name   VARCHAR(20),  
    gender      CHAR(1) NOT NULL,  
    birth_date  DATE  
);
```

- As table integrity constraints

```
CREATE TABLE actors(  
    id          CHAR(5),  
    first_name  VARCHAR(15) NOT NULL,  
    last_name   VARCHAR(20),  
    gender      CHAR(1) NOT NULL,  
    birth_date  DATE,  
    PRIMARY KEY (id)  
);
```

CREATING TABLES: SPECIFYING INTEGRITY CONSTRAINTS

ALTERNATE KEYS (UNIQUE)

- As column integrity constraints

```
CREATE TABLE movies(  
    id        CHAR(5),  
    name      VARCHAR(40) NOT NULL UNIQUE,  
    year      YEAR(4) NOT NULL,  
    rank      DECIMAL(3,1),  
    PRIMARY KEY (id)  
);
```

- As table integrity constraints

```
CREATE TABLE movies(  
    id        CHAR(5),  
    name      VARCHAR(40) NOT NULL,  
    year      YEAR(4) NOT NULL,  
    rank      DECIMAL(3,1),  
    PRIMARY KEY (id),  
    UNIQUE (name)  
);
```

CREATING TABLES: SPECIFYING INTEGRITY CONSTRAINTS

FOREIGN KEYS

- Foreign keys can be used only for tables that are created with the storage engine InnoDB (the others do not support foreign keys)

```
CREATE TABLE roles(  
    actor_id      CHAR(5),  
    movie_id     CHAR(5),  
    role         VARCHAR(40) NOT NULL,  
    salary        INTEGER,  
    PRIMARY KEY (actor_id,movie_id),  
    FOREIGN KEY (actor_id) REFERENCES actors(id),  
    FOREIGN KEY (movie_id) REFERENCES movies(id)  
)engine=InnoDB;
```


CREATING TABLES: SPECIFYING INTEGRITY CONSTRAINTS

REFERENCING ACTION

- Two statements: **ON UPDATE**, **ON DELETE**
- Five possible actions: **CASCADE**, **RESTRICT**, **SET NULL**, **NO ACTION**, **SET DEFAULT**

```
CREATE TABLE roles(  
    actor_id      CHAR(5),  
    movie_id     CHAR(5),  
    role         VARCHAR(40) NOT NULL,  
    salary       INTEGER,  
    PRIMARY KEY (actor_id,movie_id),  
    FOREIGN KEY (actor_id) REFERENCES actors(id) ON UPDATE ... ON DELETE ...,  
    FOREIGN KEY (movie_id) REFERENCES movies(id) ON UPDATE ... ON DELETE ...  
)engine=InnoDB;
```

CREATING TABLES: SPECIFYING INTEGRITY CONSTRAINTS

CHECK INTEGRITY CONSTRAINTS

- As column integrity constraints

```
CREATE TABLE actors(  
    id            CHAR(5),  
    first_name    VARCHAR(15) NOT NULL,  
    last_name     VARCHAR(20),  
    gender        CHAR(1) NOT NULL CHECK (gender IN ('M','F')),  
    birth_date    DATE,  
    PRIMARY KEY (id)  
);
```

CREATING TABLES: SPECIFYING INTEGRITY CONSTRAINTS

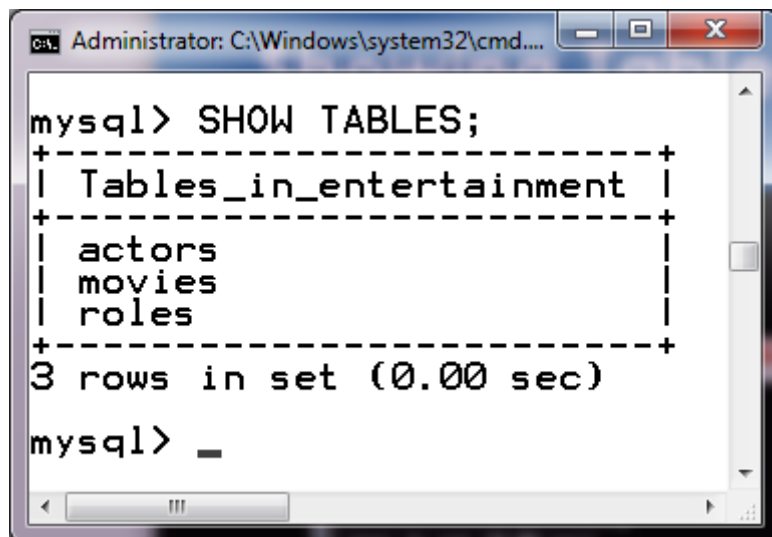
CHECK INTEGRITY CONSTRAINTS

- As table integrity constraints

```
CREATE TABLE actors(  
    id            CHAR(5),  
    first_name    VARCHAR(15) NOT NULL,  
    last_name     VARCHAR(20),  
    gender        CHAR(1) NOT NULL,  
    birth_date    DATE,  
    PRIMARY KEY (id),  
    CHECK (gender IN ('M', 'F'))  
);
```

SHOWING TABLES

SHOW TABLES;



The screenshot shows a Windows command prompt window titled "Administrator: C:\Windows\system32\cmd...". Inside the window, the command "mysql> SHOW TABLES;" has been entered. The output is displayed in a table format with dashed lines around the table name and its contents. The table has one column and three rows: "Tables_in_entertainment", "actors", "movies", and "roles". Below the table, it says "3 rows in set (0.00 sec)". The prompt "mysql> _" is visible at the bottom.

```
mysql> SHOW TABLES;
+-----+
| Tables_in_entertainment |
+-----+
| actors                  |
| movies                  |
| roles                    |
+-----+
3 rows in set (0.00 sec)

mysql> _
```

POPULATING TABLES WITH DATA

- Use the INSERT statement to add rows to an existing table

```
INSERT INTO table_name (column_name_1,column_name_2,...)  
VALUES (column_value_1,column_value_2,...);
```

- You do not have to specify column names if a value is specified for all columns of the table concerned

```
INSERT INTO table_name  
VALUES (column_value_1,column_value_2,...);
```

- If column names are omitted, MySQL assumes that the order in which the values are entered is the same as the default sequence of the columns

POPULATING TABLES WITH DATA

- Examples

```
INSERT INTO movies (id, name, year, rank)
VALUES ('M0002', 'The Hunger Games', 2012, 7.3);
```

```
INSERT INTO movies
VALUES ('M0005', 'Tooth Fairy', 2010, NULL);
```

```
INSERT INTO movies (year, name, id)
VALUES (2004, 'National Treasure', 'M0003');
```

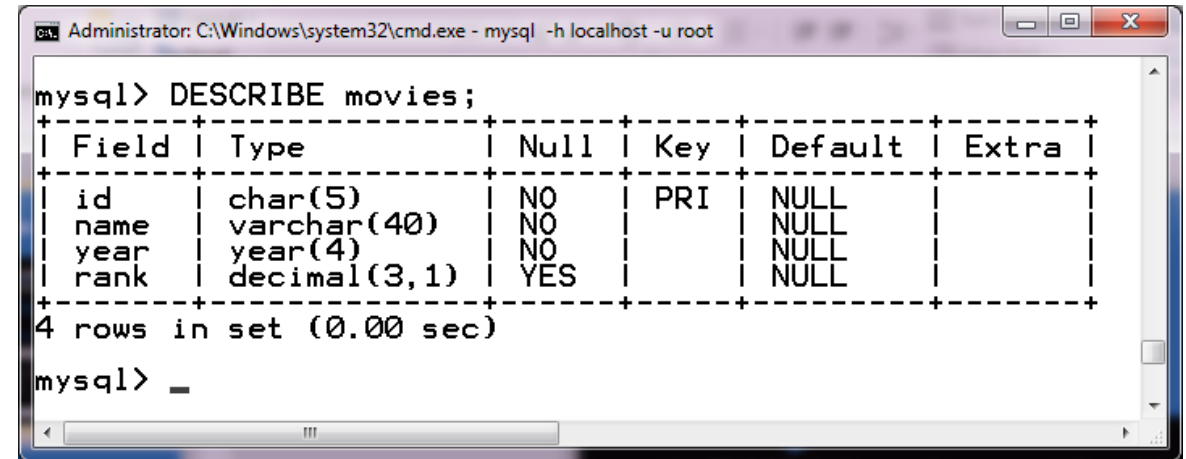
```
INSERT INTO actors
VALUES ('A0005', 'Josh', 'Hutcherson', 'M', '1992-10-12'),
      ('A0008', 'Vanessa', 'Hudgens', 'F', '1988-12-14');
```

QUERYING TABLES

- SELECT statements are used to retrieve data from tables

```
SELECT column_name_1,column_name_2,...  
FROM table_name;
```

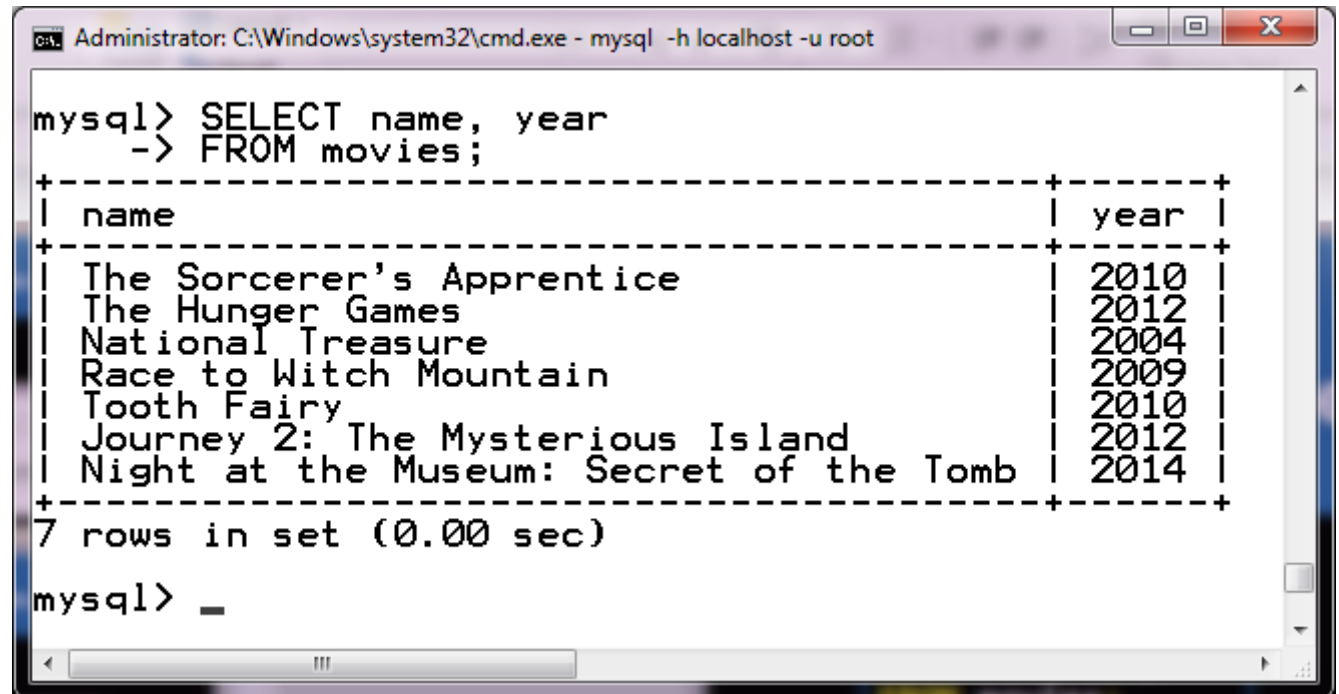
```
SELECT *  
FROM table_name;
```



```
Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root  
  
mysql> DESCRIBE movies;  
+-----+-----+-----+-----+-----+-----+  
| Field | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| id    | char(5)       | NO   | PRI | NULL    |       |  
| name  | varchar(40)   | NO   |     | NULL    |       |  
| year  | year(4)       | NO   |     | NULL    |       |  
| rank  | decimal(3,1)  | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
4 rows in set (0.00 sec)  
  
mysql> _
```

QUERYING TABLES

```
SELECT name, year  
FROM movies;
```



The screenshot shows a Windows command prompt window titled "Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root". The prompt is "mysql>". The user has entered the query "SELECT name, year FROM movies;". The output is a table with two columns: "name" and "year". The table contains seven rows of data. Below the table, it says "7 rows in set (0.00 sec)". The prompt is now "mysql> _".

```
mysql> SELECT name, year  
-> FROM movies;  
+-----+-----+  
| name                                     | year |  
+-----+-----+  
| The Sorcerer's Apprentice               | 2010  |  
| The Hunger Games                       | 2012  |  
| National Treasure                      | 2004  |  
| Race to Witch Mountain                 | 2009  |  
| Tooth Fairy                           | 2010  |  
| Journey 2: The Mysterious Island        | 2012  |  
| Night at the Museum: Secret of the Tomb | 2014  |  
+-----+-----+  
7 rows in set (0.00 sec)  
mysql> _
```


QUERYING TABLES

SELECT *
FROM movies;

```
Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root

mysql> SELECT *
-> FROM movies;

+----+-----+-----+-----+
| id  | name                                     | year | rank |
+----+-----+-----+-----+
| M0001 | The Sorcerer's Apprentice             | 2010 | 6.2  |
| M0002 | The Hunger Games                      | 2012 | 7.3  |
| M0003 | National Treasure                     | 2004 | 6.9  |
| M0004 | Race to Witch Mountain               | 2009 | 5.7  |
| M0005 | Tooth Fairy                           | 2010 | 5.0  |
| M0006 | Journey 2: The Mysterious Island      | 2012 | 5.8  |
| M0007 | Night at the Museum: Secret of the Tomb | 2014 | 6.6  |
+----+-----+-----+-----+
7 rows in set (0.00 sec)

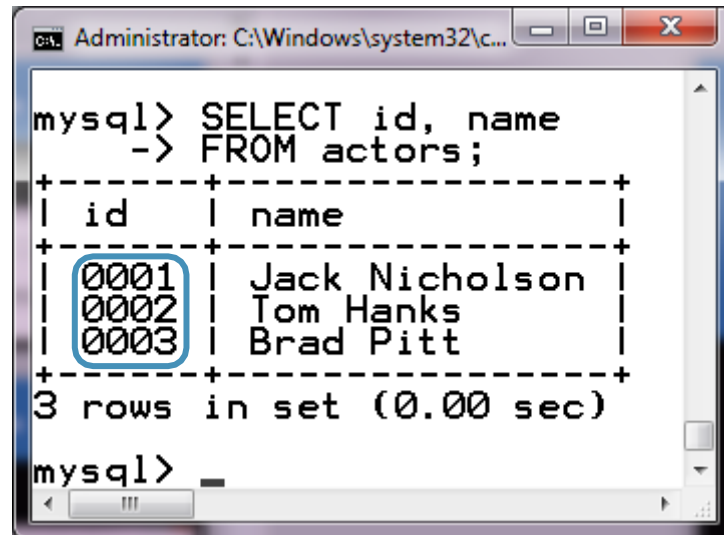
mysql> _
```

DATA TYPE OPTION: ZEROFILL

- Adding ZEROFILL affects the way in which numeric values are presented
- If the width of a numeric value is smaller than that of the maximum width allowed, the value is filled in the front with zeroes

```
CREATE TABLE actors(  
    id      INTEGER(4) ZEROFILL,  
    name    VARCHAR(25)  
);
```

```
INSERT INTO actors  
VALUES (1, 'Jack Nicholson'),  
       (2, 'Tom Hanks'),  
       (3, 'Brad Pitt');
```



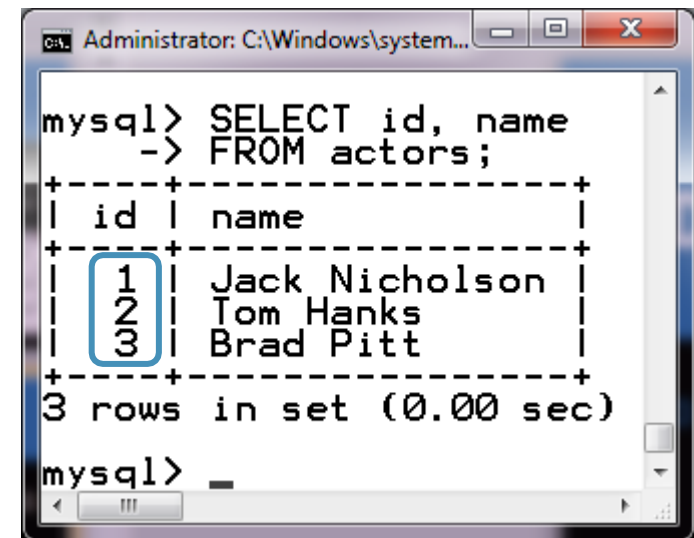
```
Administrator: C:\Windows\system32\c...  
mysql> SELECT id, name  
-> FROM actors;  
+----+-----+  
| id | name      |  
+----+-----+  
| 0001 | Jack Nicholson |  
| 0002 | Tom Hanks      |  
| 0003 | Brad Pitt      |  
+----+-----+  
3 rows in set (0.00 sec)  
mysql>
```

DATA TYPE OPTION: AUTO_INCREMENT

- If a NULL value is specified for the AUTO_INCREMENT column in the INSERT statement, or if no value is specified, MySQL determines what the next sequence number should be

```
CREATE TABLE actors(  
    id      INTEGER AUTO_INCREMENT PRIMARY KEY,  
    name    VARCHAR(25)  
);
```

```
INSERT INTO actors (name)  
VALUES ('Jack Nicholson'),  
      ('Tom Hanks'),  
      ('Brad Pitt');
```



The screenshot shows a MySQL command window titled "Administrator: C:\Windows\system...". The command prompt shows the following SQL query and its result:

```
mysql> SELECT id, name  
-> FROM actors;
```

id	name
1	Jack Nicholson
2	Tom Hanks
3	Brad Pitt

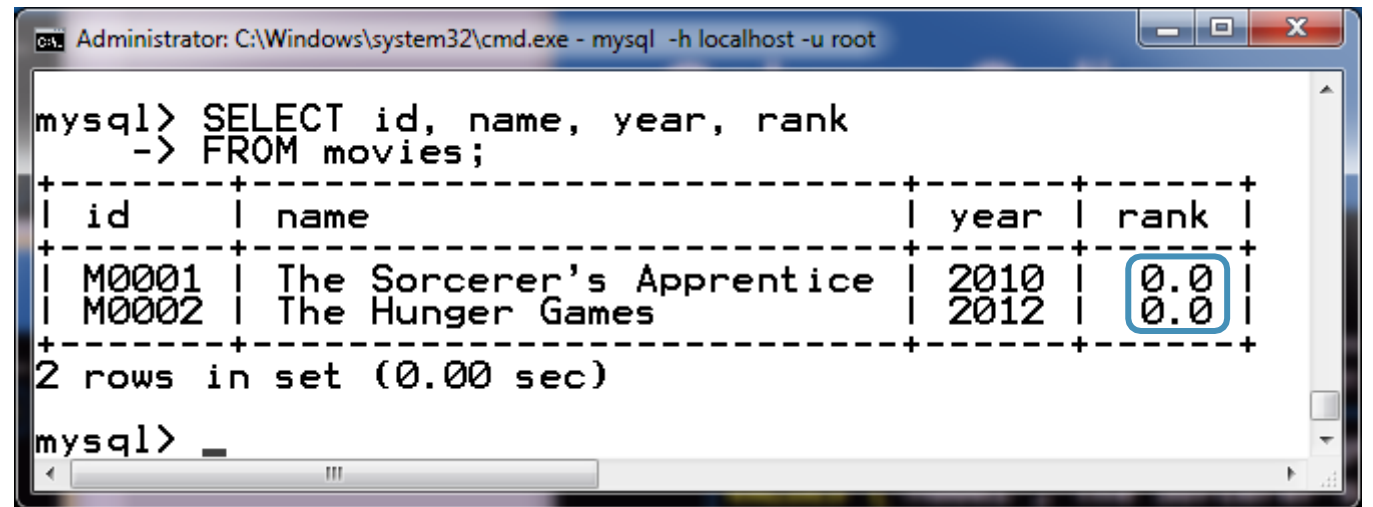
Below the table, the output indicates "3 rows in set (0.00 sec)". The command prompt then shows "mysql> " followed by a cursor.

COLUMN OPTION: DEFAULT

- DEFAULT value is used when a new row is added to a table and no value has been specified for that column

```
CREATE TABLE movies(  
  id      CHAR(5) PRIMARY KEY,  
  name    VARCHAR(40) NOT NULL,  
  year    YEAR(4) NOT NULL,  
  rank    DECIMAL(3,1) DEFAULT 0  
);
```

```
INSERT INTO movies (id, name, year)  
VALUES ('M0001', 'The Sorcerer's Apprentice', 2010),  
      ('M0002', 'The Hunger Games', 2012);
```



The screenshot shows a Windows command prompt window titled "Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root". Inside the window, the following MySQL commands and output are visible:

```
mysql> SELECT id, name, year, rank  
        -> FROM movies;
```

id	name	year	rank
M0001	The Sorcerer's Apprentice	2010	0.0
M0002	The Hunger Games	2012	0.0

2 rows in set (0.00 sec)

```
mysql> _
```

The output shows two rows of data. The 'rank' column for both rows contains the value '0.0', which is highlighted with a blue box in the original image, demonstrating the default value being applied.

UPDATING ROWS

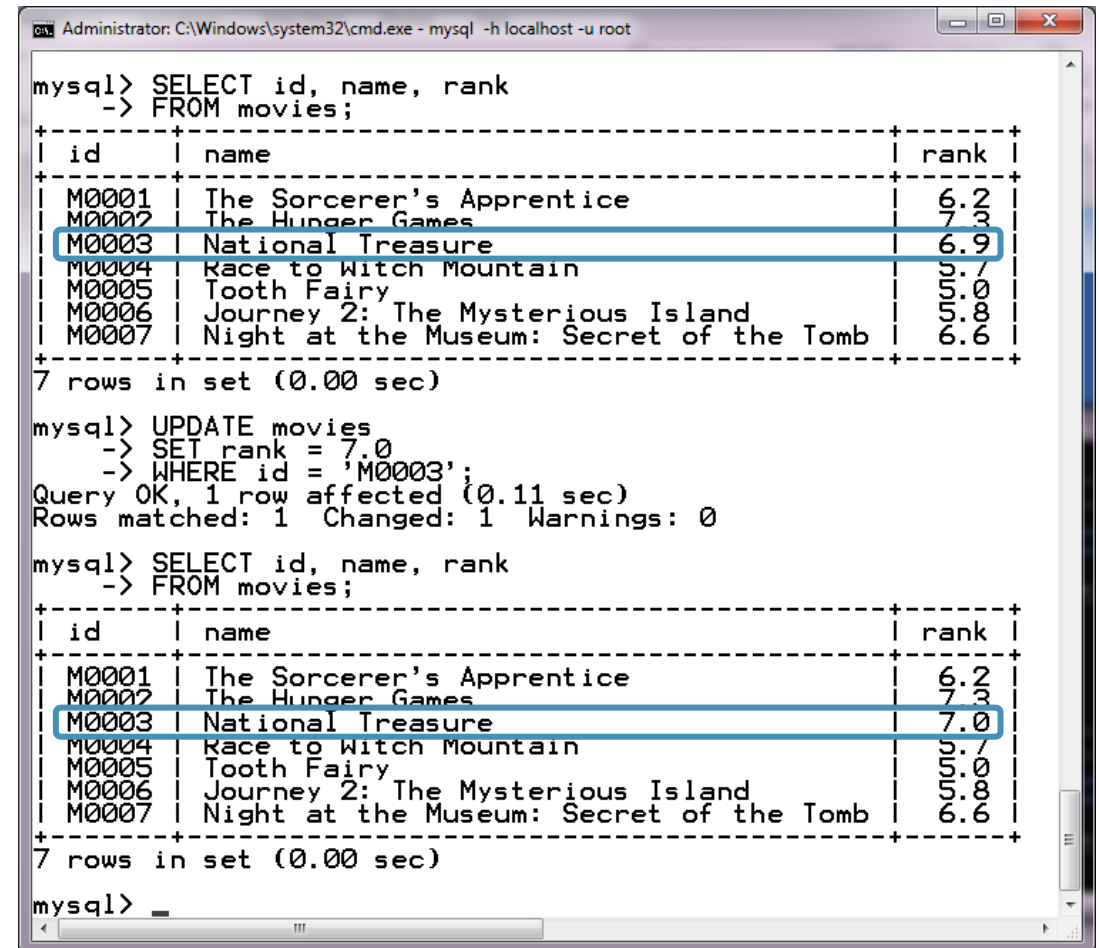
- UPDATE statement is used to change values in rows

```
UPDATE table_name  
SET column_name = new_column_value  
WHERE condition;
```

UPDATING ROWS

```
UPDATE movies  
SET rank = 7.0  
WHERE id = 'M0003';
```

```
UPDATE movies  
SET rank = 7.0  
WHERE name = 'National Treasure';
```



The screenshot shows a Windows command prompt window titled "Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root". It displays the following MySQL commands and their outputs:

```
mysql> SELECT id, name, rank  
-> FROM movies;
```

id	name	rank
M0001	The Sorcerer's Apprentice	6.2
M0002	The Hunger Games	7.3
M0003	National Treasure	6.9
M0004	Race to Witch Mountain	5.7
M0005	Tooth Fairy	5.0
M0006	Journey 2: The Mysterious Island	5.8
M0007	Night at the Museum: Secret of the Tomb	6.6

7 rows in set (0.00 sec)

```
mysql> UPDATE movies  
-> SET rank = 7.0  
-> WHERE id = 'M0003';  
Query OK, 1 row affected (0.11 sec)  
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> SELECT id, name, rank  
-> FROM movies;
```

id	name	rank
M0001	The Sorcerer's Apprentice	6.2
M0002	The Hunger Games	7.3
M0003	National Treasure	7.0
M0004	Race to Witch Mountain	5.7
M0005	Tooth Fairy	5.0
M0006	Journey 2: The Mysterious Island	5.8
M0007	Night at the Museum: Secret of the Tomb	6.6

7 rows in set (0.00 sec)

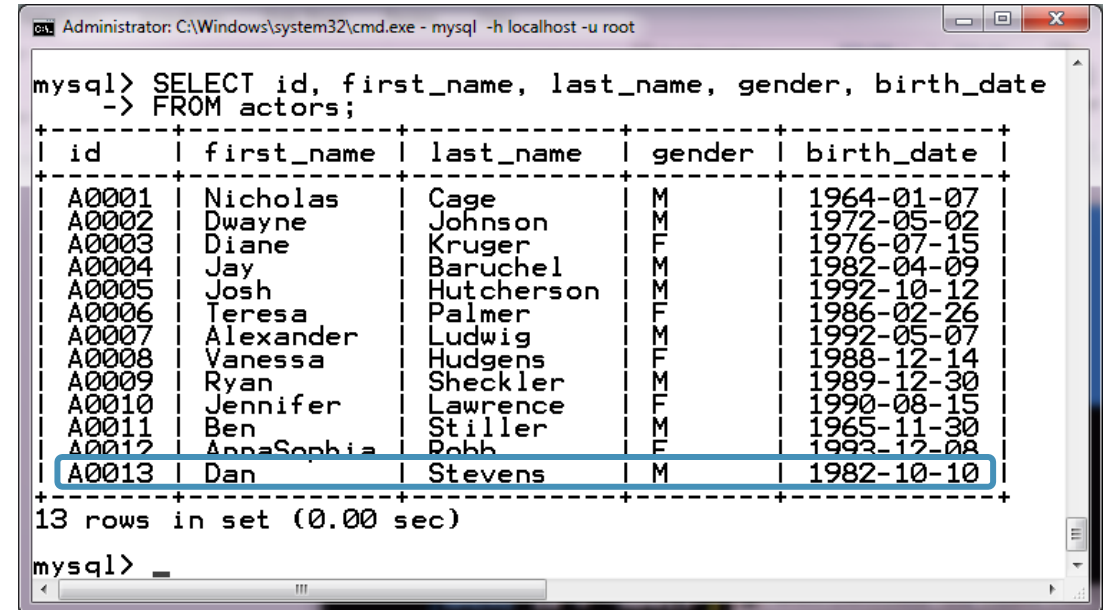
```
mysql> _
```

UPDATING ROWS

UPDATE actors

SET first_name = 'Daniel',
birth_date = '1982-12-12'

WHERE id = 'A0013';



The screenshot shows a Windows command prompt window titled "Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root". The prompt shows the command `mysql> SELECT id, first_name, last_name, gender, birth_date -> FROM actors;` and the resulting table of 13 rows. The row for id 'A0013' is highlighted with a blue box.

id	first_name	last_name	gender	birth_date
A0001	Nicholas	Cage	M	1964-01-07
A0002	Dwayne	Johnson	M	1972-05-02
A0003	Diane	Kruger	F	1976-07-15
A0004	Jay	Baruchel	M	1982-04-09
A0005	Josh	Hutcherson	M	1992-10-12
A0006	Teresa	Palmer	F	1986-02-26
A0007	Alexander	Ludwig	M	1992-05-07
A0008	Vanessa	Hudgens	F	1988-12-14
A0009	Ryan	Sheckler	M	1989-12-30
A0010	Jennifer	Lawrence	F	1990-08-15
A0011	Ben	Stiller	M	1965-11-30
A0012	AnnaSophia	Robb	F	1993-12-08
A0013	Dan	Stevens	M	1982-10-10

13 rows in set (0.00 sec)

mysql> _

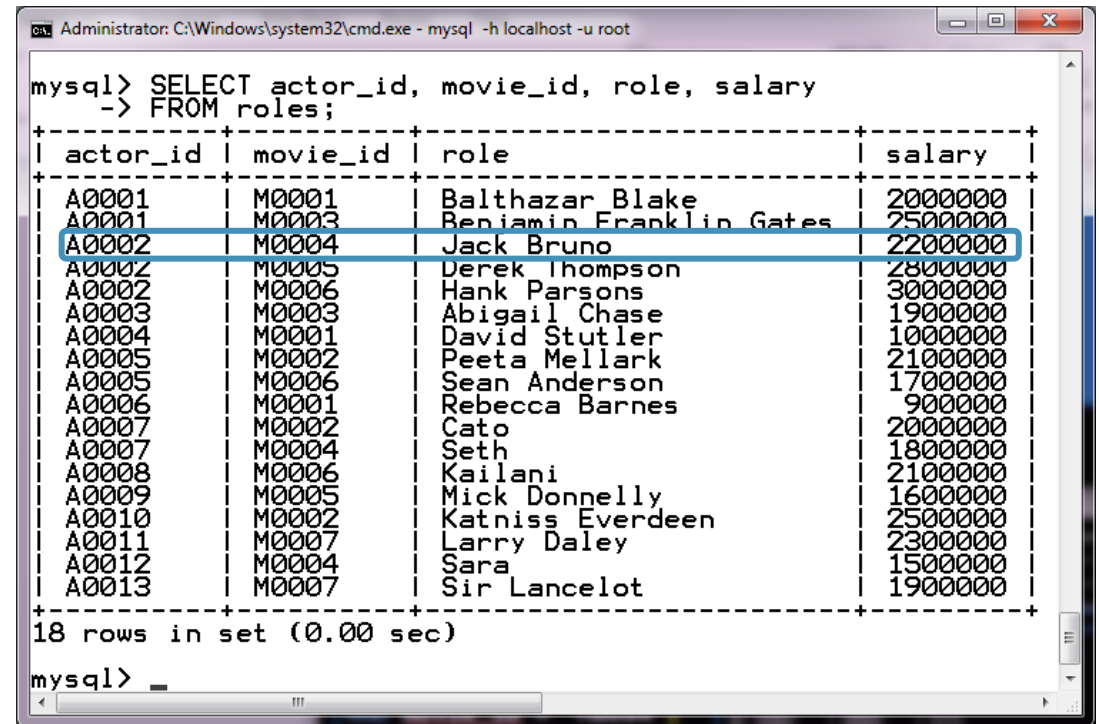
UPDATING ROWS

UPDATE roles

SET salary = 2400000

WHERE actor_id = 'A0002'

AND movie_id = 'M0004';



actor_id	movie_id	role	salary
A0001	M0001	Balthazar Blake	2000000
A0001	M0003	Benjamin Franklin Gates	2500000
A0002	M0004	Jack Bruno	2200000
A0002	M0005	Derek Thompson	2800000
A0002	M0006	Hank Parsons	3000000
A0003	M0003	Abigail Chase	1900000
A0004	M0001	David Stutler	1000000
A0005	M0002	Peeta Mellark	2100000
A0005	M0006	Sean Anderson	1700000
A0006	M0001	Rebecca Barnes	900000
A0007	M0002	Cato	2000000
A0007	M0004	Seth	1800000
A0008	M0006	Kailani	2100000
A0009	M0005	Mick Donnelly	1600000
A0010	M0002	Katniss Everdeen	2500000
A0011	M0007	Larry Daley	2300000
A0012	M0004	Sara	1500000
A0013	M0007	Sir Lancelot	1900000

18 rows in set (0.00 sec)

mysql> _

UPDATING ROWS

UPDATE roles

SET salary = salary + 200000;

```
Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root

mysql> SELECT actor_id, movie_id, role, salary
-> FROM roles;
```

actor_id	movie_id	role	salary
A0001	M0001	Balthazar Blake	2000000
A0001	M0003	Benjamin Franklin Gates	2500000
A0002	M0004	Jack Bruno	2200000
A0002	M0005	Derek Thompson	2800000
A0002	M0006	Hank Parsons	3000000
A0003	M0003	Abigail Chase	1900000
A0004	M0001	David Stutler	1000000
A0005	M0002	Peeta Mellark	2100000
A0005	M0006	Sean Anderson	1700000
A0006	M0001	Rebecca Barnes	900000
A0007	M0002	Cato	2000000
A0007	M0004	Seth	1800000
A0008	M0006	Kailani	2100000
A0009	M0005	Mick Donnelly	1600000
A0010	M0002	Katniss Everdeen	2500000
A0011	M0007	Larry Daley	2300000
A0012	M0004	Sara	1500000
A0013	M0007	Sir Lancelot	1900000

```
18 rows in set (0.00 sec)

mysql>
```

DELETING ROWS

- DELETE statement is used to remove complete rows from a table

```
DELETE  
FROM table_name  
WHERE condition;
```

DELETING ROWS

DELETE

FROM movies

WHERE rank < 6.0;

```
Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root

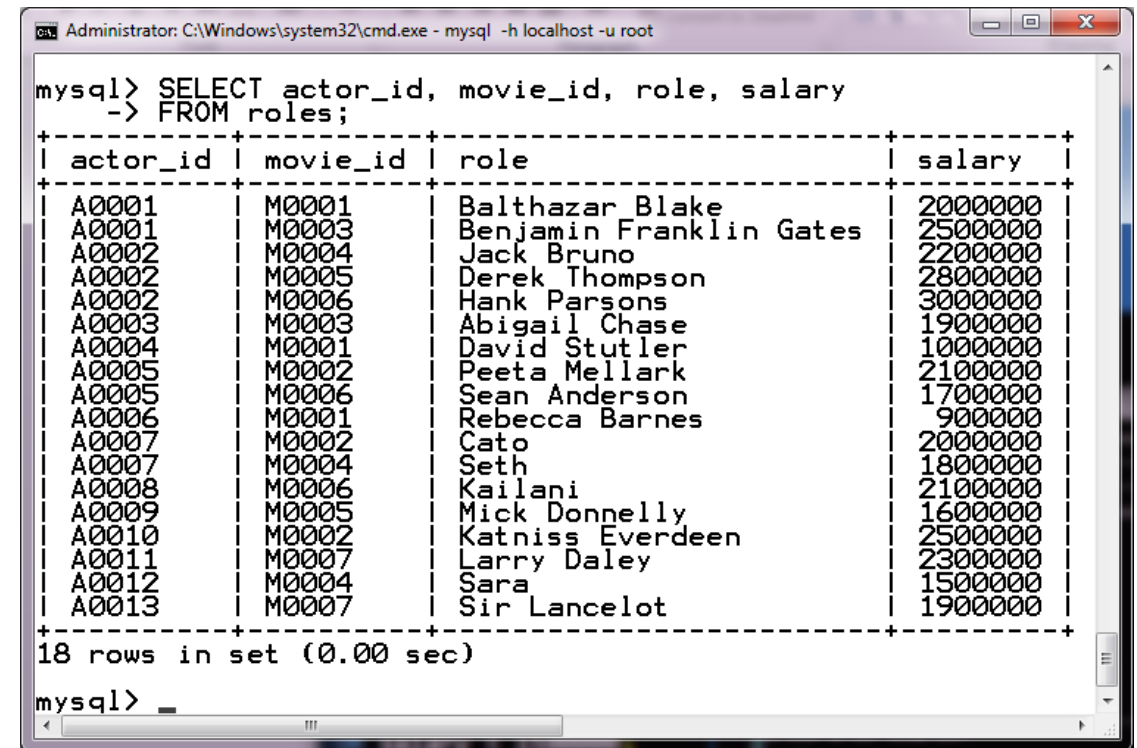
mysql> SELECT id, name, year, rank
-> FROM movies;
+-----+-----+-----+-----+
| id      | name                                     | year | rank |
+-----+-----+-----+-----+
| M0001   | The Sorcerer's Apprentice               | 2010  | 6.2  |
| M0002   | The Hunger Games                       | 2012  | 7.3  |
| M0003   | National Treasure                      | 2004  | 6.9  |
| M0004   | Race to Witch Mountain                 | 2009  | 5.7  |
| M0005   | Tooth Fairy                           | 2010  | 5.0  |
| M0006   | Journey 2: The Mysterious Island       | 2012  | 5.8  |
| M0007   | Night at the Museum: Secret of the Tomb | 2014  | 6.6  |
+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> _
```

DELETING ROWS

```
DELETE
FROM roles
WHERE actor_id = 'A0002'
AND salary < 1500000;
```

```
DELETE
FROM roles
WHERE actor_id = 'A0002'
OR salary < 1500000;
```



The screenshot shows a Windows command prompt window titled "Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root". The prompt is "mysql>". The user has entered the command "SELECT actor_id, movie_id, role, salary -> FROM roles;". The output is a table with 18 rows and 4 columns: actor_id, movie_id, role, and salary. The data is as follows:

actor_id	movie_id	role	salary
A0001	M0001	Balthazar Blake	2000000
A0001	M0003	Benjamin Franklin Gates	2500000
A0002	M0004	Jack Bruno	2200000
A0002	M0005	Derek Thompson	2800000
A0002	M0006	Hank Parsons	3000000
A0003	M0003	Abigail Chase	1900000
A0004	M0001	David Stutler	1000000
A0005	M0002	Peeta Mellark	2100000
A0005	M0006	Sean Anderson	1700000
A0006	M0001	Rebecca Barnes	900000
A0007	M0002	Cato	2000000
A0007	M0004	Seth	1800000
A0008	M0006	Kailani	2100000
A0009	M0005	Mick Donnelly	1600000
A0010	M0002	Katniss Everdeen	2500000
A0011	M0007	Larry Daley	2300000
A0012	M0004	Sara	1500000
A0013	M0007	Sir Lancelot	1900000

Below the table, it says "18 rows in set (0.00 sec)". The prompt is now "mysql> _".

DELETING ROWS

DELETE
FROM roles;

```
Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root
mysql> SELECT actor_id, movie_id, role, salary
-> FROM roles;
```

actor_id	movie_id	role	salary
A0001	M0001	Balthazar Blake	2000000
A0001	M0003	Benjamin Franklin Gates	2500000
A0002	M0004	Jack Bruno	2200000
A0002	M0005	Derek Thompson	2800000
A0002	M0006	Hank Parsons	3000000
A0003	M0003	Abigail Chase	1900000
A0004	M0001	David Stutler	1000000
A0005	M0002	Peeta Mellark	2100000
A0005	M0006	Sean Anderson	1700000
A0006	M0001	Rebecca Barnes	900000
A0007	M0002	Cato	2000000
A0007	M0004	Seth	1800000
A0008	M0006	Kailani	2100000
A0009	M0005	Mick Donnelly	1600000
A0010	M0002	Katniss Everdeen	2500000
A0011	M0007	Larry Daley	2300000
A0012	M0004	Sara	1500000
A0013	M0007	Sir Lancelot	1900000

```
18 rows in set (0.00 sec)
mysql> _
```

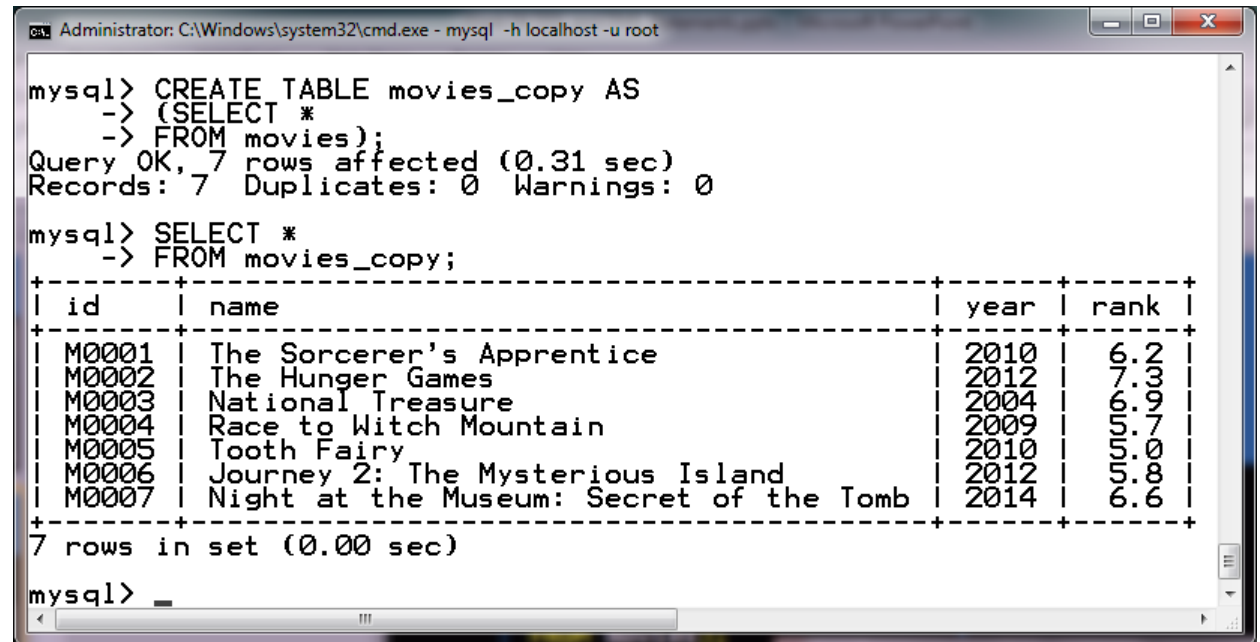
COPYING TABLES

- CREATE TABLE statement is also possible to create a new table that is based on an existing table
- The specifications and / or the contents of the existing tables are used to create the new table and possibly fill it as well

```
CREATE TABLE table_name AS  
(SELECT statement);
```

COPYING TABLES

```
CREATE TABLE movies_copy AS  
(SELECT *  
FROM movies);
```



The screenshot shows a Windows command prompt window titled "Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root". The MySQL command line interface is active, showing the following commands and output:

```
mysql> CREATE TABLE movies_copy AS  
-> (SELECT *  
-> FROM movies);  
Query OK, 7 rows affected (0.31 sec)  
Records: 7 Duplicates: 0 Warnings: 0  
  
mysql> SELECT *  
-> FROM movies_copy;
```

The output of the second query is displayed in a table format with dashed line borders:

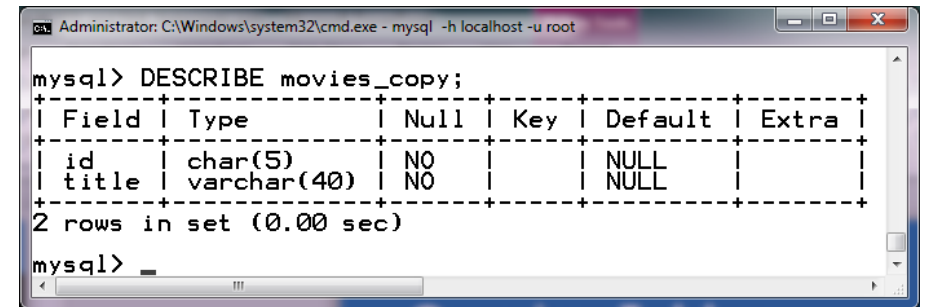
id	name	year	rank
M0001	The Sorcerer's Apprentice	2010	6.2
M0002	The Hunger Games	2012	7.3
M0003	National Treasure	2004	6.9
M0004	Race to Witch Mountain	2009	5.7
M0005	Tooth Fairy	2010	5.0
M0006	Journey 2: The Mysterious Island	2012	5.8
M0007	Night at the Museum: Secret of the Tomb	2014	6.6

Below the table, the output continues with:

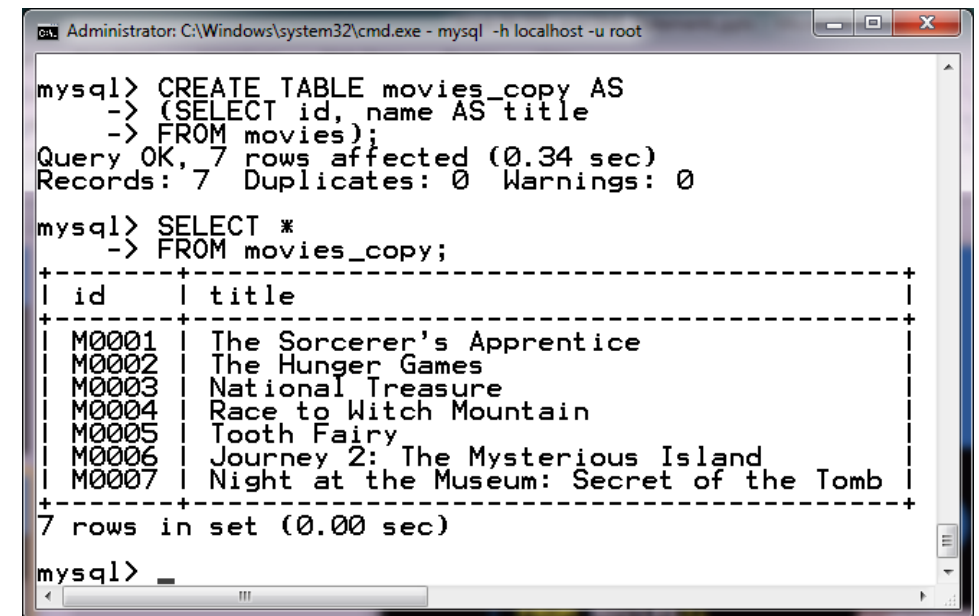
```
7 rows in set (0.00 sec)  
  
mysql> _
```

COPYING TABLES

```
CREATE TABLE movies_copy AS  
(SELECT id, name AS title  
FROM movies);
```



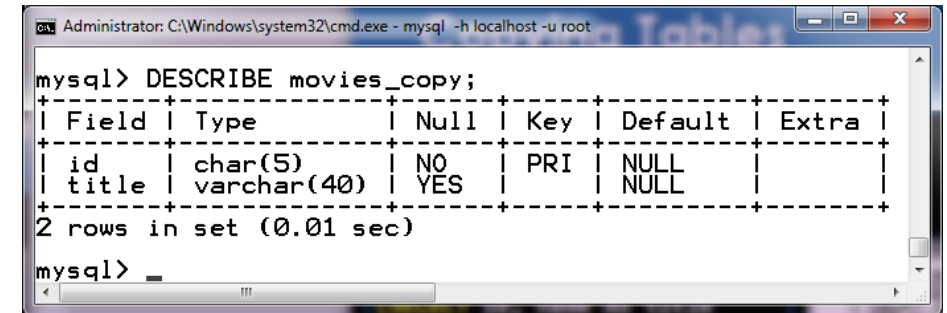
```
Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root  
  
mysql> DESCRIBE movies_copy;  
+-----+-----+-----+-----+-----+-----+  
| Field | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| id    | char(5)       | NO   |     | NULL    |       |  
| title | varchar(40)   | NO   |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
2 rows in set (0.00 sec)  
  
mysql> _
```



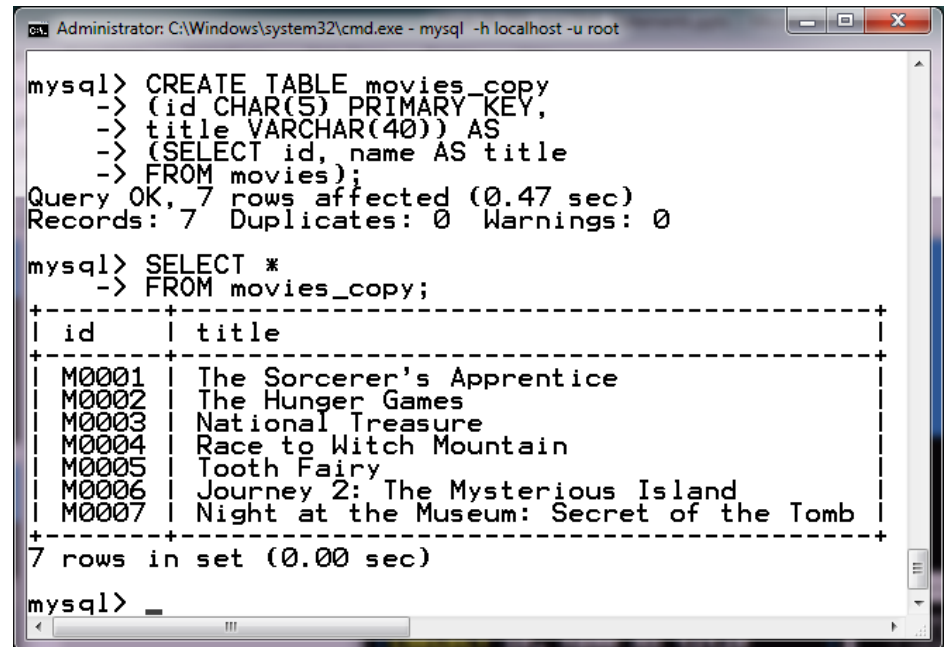
```
Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root  
  
mysql> CREATE TABLE movies_copy AS  
-> (SELECT id, name AS title  
-> FROM movies);  
Query OK, 7 rows affected (0.34 sec)  
Records: 7 Duplicates: 0 Warnings: 0  
  
mysql> SELECT *  
-> FROM movies_copy;  
+-----+-----+  
| id    | title                                     |  
+-----+-----+  
| M0001 | The Sorcerer's Apprentice               |  
| M0002 | The Hunger Games                        |  
| M0003 | National Treasure                       |  
| M0004 | Race to Witch Mountain                 |  
| M0005 | Tooth Fairy                             |  
| M0006 | Journey 2: The Mysterious Island        |  
| M0007 | Night at the Museum: Secret of the Tomb |  
+-----+-----+  
7 rows in set (0.00 sec)  
  
mysql> _
```


COPYING TABLES

```
CREATE TABLE movies_copy  
(id      CHAR(5) PRIMARY KEY,  
 title  VARCHAR(40)) AS  
(SELECT id, name AS title  
 FROM movies);
```



```
Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root  
  
mysql> DESCRIBE movies_copy;  
+-----+-----+-----+-----+-----+-----+  
| Field | Type   | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| id    | char(5) | NO   | PRI | NULL    |       |  
| title | varchar(40) | YES |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
2 rows in set (0.01 sec)  
  
mysql> _
```



```
Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root  
  
mysql> CREATE TABLE movies_copy  
-> (id CHAR(5) PRIMARY KEY,  
-> title VARCHAR(40)) AS  
-> (SELECT id, name AS title  
-> FROM movies);  
Query OK, 7 rows affected (0.47 sec)  
Records: 7  Duplicates: 0  Warnings: 0  
  
mysql> SELECT *  
-> FROM movies_copy;  
+-----+-----+  
| id | title |  
+-----+-----+  
| M0001 | The Sorcerer's Apprentice |  
| M0002 | The Hunger Games |  
| M0003 | National Treasure |  
| M0004 | Race to Witch Mountain |  
| M0005 | Tooth Fairy |  
| M0006 | Journey 2: The Mysterious Island |  
| M0007 | Night at the Museum: Secret of the Tomb |  
+-----+-----+  
7 rows in set (0.00 sec)  
  
mysql> _
```

COPYING TABLES

CREATE TABLE movies_copy
LIKE movies;

```
Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root

mysql> SHOW TABLES;
+-----+
| Tables_in_entertainment |
+-----+
| actors                  |
| movies                  |
| roles                   |
+-----+
3 rows in set (0.00 sec)

mysql> CREATE TABLE movies_copy LIKE movies;
Query OK, 0 rows affected (0.27 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_entertainment |
+-----+
| actors                  |
| movies                  |
| movies_copy             |
| roles                   |
+-----+
4 rows in set (0.00 sec)

mysql> DESCRIBE movies_copy;
+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+
| id     | char(5)       | NO   | PRI | NULL    |       |
| name   | varchar(40)   | NO   |     | NULL    |       |
| year   | year(4)       | NO   |     | NULL    |       |
| rank   | decimal(3,1)  | YES  |     | NULL    |       |
+-----+
4 rows in set (0.00 sec)

mysql> SELECT *
-> FROM movies_copy;
Empty set (0.00 sec)

mysql>
```

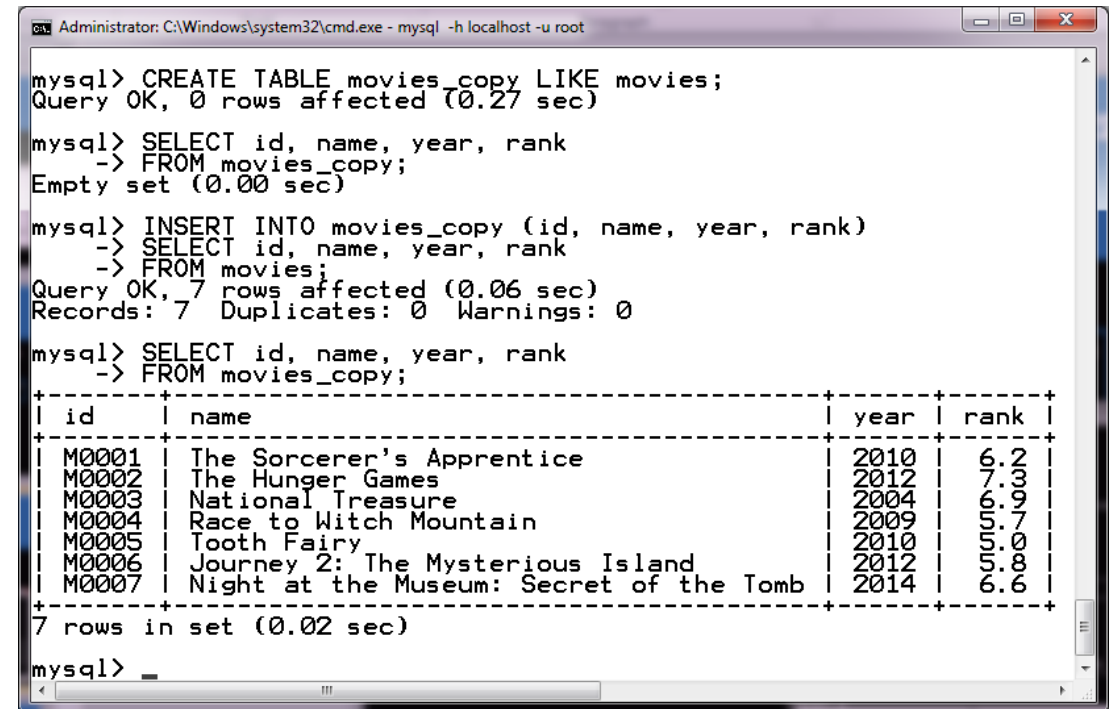
POPULATING A TABLE WITH ROWS FROM ANOTHER TABLE

- With the INSERT statement, we can fill a table with rows from another table / other tables
- The data is copied from one table to another

```
INSERT INTO table_name (column_name_1,column_name_2,...)  
SELECT statement;
```

POPULATING A TABLE WITH ROWS FROM ANOTHER TABLE

```
INSERT INTO movies_copy (id, name, year, rank)
SELECT id, name, year, rank
FROM movies;
```



```
Administrator: C:\Windows\system32\cmd.exe - mysql -h localhost -u root

mysql> CREATE TABLE movies_copy LIKE movies;
Query OK, 0 rows affected (0.27 sec)

mysql> SELECT id, name, year, rank
-> FROM movies_copy;
Empty set (0.00 sec)

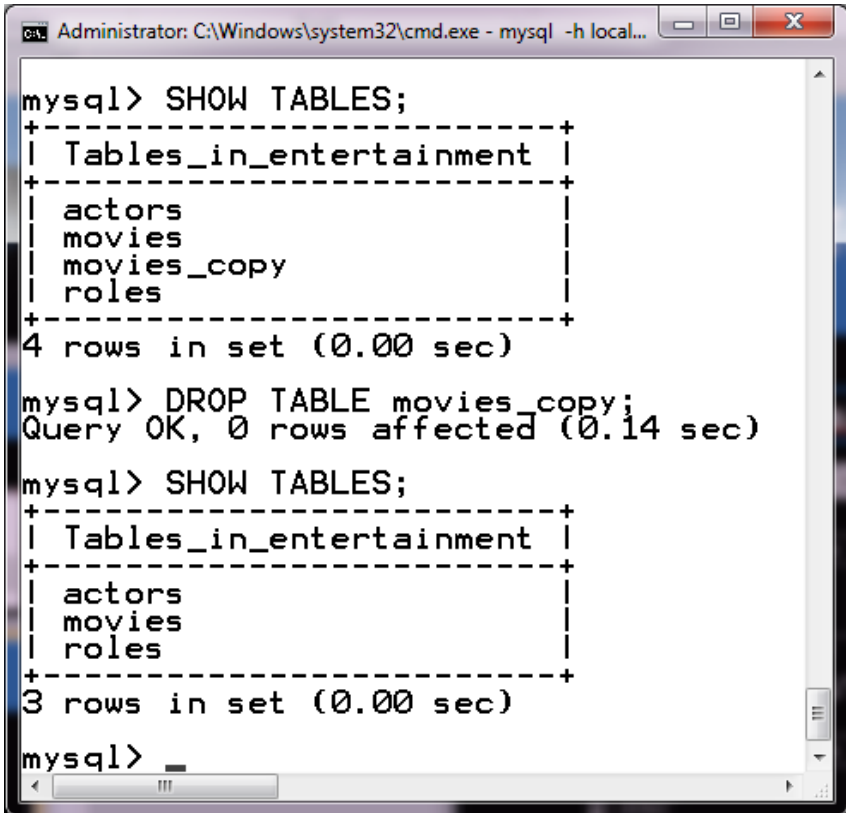
mysql> INSERT INTO movies_copy (id, name, year, rank)
-> SELECT id, name, year, rank
-> FROM movies;
Query OK, 7 rows affected (0.06 sec)
Records: 7 Duplicates: 0 Warnings: 0

mysql> SELECT id, name, year, rank
-> FROM movies_copy;
+----+-----+-----+-----+
| id  | name                                     | year | rank |
+----+-----+-----+-----+
| M0001 | The Sorcerer's Apprentice             | 2010 | 6.2  |
| M0002 | The Hunger Games                      | 2012 | 7.3  |
| M0003 | National Treasure                    | 2004 | 6.9  |
| M0004 | Race to Witch Mountain               | 2009 | 5.7  |
| M0005 | Tooth Fairy                          | 2010 | 5.0  |
| M0006 | Journey 2: The Mysterious Island      | 2012 | 5.8  |
| M0007 | Night at the Museum: Secret of the Tomb | 2014 | 6.6  |
+----+-----+-----+-----+
7 rows in set (0.02 sec)

mysql> _
```

DELETING TABLES

DROP TABLE table_name;



The screenshot shows a Windows command prompt window titled "Administrator: C:\Windows\system32\cmd.exe - mysql -h local...". The window contains the following text:

```
mysql> SHOW TABLES;
+-----+
| Tables_in_entertainment |
+-----+
| actors                   |
| movies                   |
| movies_copy              |
| roles                    |
+-----+
4 rows in set (0.00 sec)

mysql> DROP TABLE movies_copy;
Query OK, 0 rows affected (0.14 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_entertainment |
+-----+
| actors                   |
| movies                   |
| roles                    |
+-----+
3 rows in set (0.00 sec)

mysql> _
```

REFERENCES

- Elmasri, Ramez and Shamkant B. Navathe (2016), Fundamentals of Database Systems, 7th edition, Addison Wesley.
- Rick F.van der Lans (2007), SQL for MySQL Developers, Addison Wesley.
- Connolly, Thomas and Carolyn Begg (2015), Database Systems A Practical Approach to Design, Implementation, and Management, 6th edition, Pearson Education, ISBN 13: 978-1-292-06118-4, Essex, England.

NEXT WEEK: SELECT STATEMENT 1

- SELECT
- FROM
- WHERE
- Operators
- Aggregation Functions

```
SELECT *  
FROM WORLD  
WHERE SOMEONE  
LIKE '%YOU%';
```

```
EMPTY SET (0.00 SEC)
```

Vision

To become an **outstanding** undergraduate Computer Science program that produces **international-minded** graduates who are **competent** in software engineering and have **entrepreneurial spirit** and **noble character**.



Mission

1. To conduct studies with the best technology and curriculum, supported by professional lecturer
2. To conduct research in Informatics to promote science and technology
3. To deliver science-and-technology-based society services to implement science and technology